



DineshTrade

EC2 Server Setup & Deployment Guide

AWS Mumbai (ap-south-1) · Ubuntu 24.04 LTS · May 2026

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Server Reference Card

Instance ID	i-0261e0d98f83e1458
Elastic IP (Static)	3.111.255.172
Region	Mumbai – ap-south-1
OS	Ubuntu Server 24.04 LTS
Instance Type	t3.micro (Free tier – 750 hrs/month)
Storage	20 GiB gp3
Domain	dineshtrade.online
SSH Key	dineshtrade-server.pem
SSH User	ubuntu

01

AWS Account Setup

STEP 1.1

Create AWS Account

1. Go to aws.amazon.com → Click "Create an AWS Account"
2. Enter email → set password → Account name: DineshTrade
3. Choose Personal account type
4. Enter details + credit card (verification only)
5. Phone OTP verification
6. Choose Basic Support (Free)
7. Click "Go to AWS Console"

NOTE: Account is on \$100 credit / 184 days plan. EC2 t3.micro is free for first 750 hours/month. Elastic IP is free while attached to a running instance.

STEP 1.2

Switch Region to Mumbai

Top right corner → Click region dropdown → Select "Asia Pacific (Mumbai) ap-south-1"

IMPORTANT: Always verify Mumbai is selected before any EC2 operations.

02

Launch EC2 Instance

STEP 2.1

Launch Instance Configuration

Setting	Value
Name	dineshtrade-server
AMI	Ubuntu Server 24.04 LTS (HVM) — Free tier eligible
Instance type	t3.micro — Free tier eligible
Key pair	dineshtrade-server (RSA, .pem format) — Download & save safely
Storage	20 GiB gp3 (3000 IOPS)
Security Group	dineshtrade-sg

STEP 2.2

Security Group Rules (Inbound)

Rule	Type	Protocol	Port	Source
1	SSH	TCP	22	My IP only (122.167.118.47/32)
2	HTTP	TCP	80	Anywhere (0.0.0.0/0)
3	HTTPS	TCP	443	Anywhere (0.0.0.0/0)

4	Custom TCP	TCP	3000	Anywhere (0.0.0.0/0)
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WARNING: If your home IP changes, SSH rule (port 22) will block you. Update the security group in AWS Console → EC2 → Security Groups → dineshtrade-sg → Edit inbound rules → update SSH source to new IP.

03

Elastic IP — Static IP Assignment

STEP 3.1

Allocate Elastic IP

1. EC2 Dashboard → Left sidebar → Network & Security → Elastic IPs
2. Click "Allocate Elastic IP address" (orange button)
3. Leave all defaults → Click "Allocate"
4. Your static IP: 3.111.255.172

STEP 3.2

Associate IP to Instance

1. Select the Elastic IP → Actions → Associate Elastic IP address
2. Resource type: Instance
3. Instance: i-0261e0d98f83e1458
4. Private IP: 172.31.46.156 (auto-filled)
5. Click "Associate"

COST NOTE: Elastic IP is FREE while attached to a running instance. It costs ~\$0.005/hour only if the instance is STOPPED. Solution: never stop the instance — only reboot if needed.

ZERODHA: Register 3.111.255.172 at developers.kite.trade → Profile → IP Whitelist. This IP covers all family accounts (Dinesh, Kiran, Sheela, Sonia).

04

Install Server Software

STEP 4.1

SSH Into Server

```
# Fix key permissions (run once)

chmod 400 ~/Downloads/dineshtrade-server.pem

# Connect to server

ssh -i ~/Downloads/dineshtrade-server.pem ubuntu@3.111.255.172

# Type "yes" when asked about fingerprint

# You should see: ubuntu@ip-172-31-46-156:~$
```

STEP 4.2

Install All Required Software

```
# Step 1: Update system packages

sudo apt update && sudo apt upgrade -y


# Step 2: Install Node.js v20

curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -

sudo apt install -y nodejs


# Verify: should show v20.x.x and 10.x.x

node --version && npm --version


# Step 3: Install PM2 (process manager) + Nginx (web server)

sudo npm install -g pm2 && sudo apt install -y nginx


# Step 4: Start Nginx and enable on boot

sudo systemctl start nginx && sudo systemctl enable nginx
```

What each component does:

Software	Purpose	Analogy
Node.js v20	Runtime for Next.js app	Like JVM for Java
PM2 v7	Keeps app running 24/7, auto-restart on crash/reboot	Like a Windows Service
Nginx	Web server — routes port 80/443 to app on port 3000	Like Apache/Tomcat
Certbot	Free SSL certificate (Let's Encrypt), auto-renews	Like a free SSL provider

STEP 5.1

Create Nginx Config for DineshTrade

```
# Create config file

sudo nano /etc/nginx/sites-available/dineshtrade

# Paste this content:

server {

    listen 80;

    server_name dineshtrade.online www.dineshtrade.online 3.111.255.172;

    location / {

        proxy_pass http://localhost:3000;

        proxy_http_version 1.1;

        proxy_set_header Upgrade $http_upgrade;

        proxy_set_header Connection 'upgrade';

        proxy_set_header Host $host;

        proxy_cache_bypass $http_upgrade;

    }

}

# Save: Ctrl+X → Y → Enter
```

```
# Enable the site

sudo ln -s /etc/nginx/sites-available/dineshtrade /etc/nginx/sites-enabled/

# Remove default site

sudo rm /etc/nginx/sites-enabled/default
```

```
# Test config and restart

sudo nginx -t && sudo systemctl restart nginx

# Expected output:

# nginx: the configuration file /etc/nginx/nginx.conf syntax is ok

# nginx: configuration file /etc/nginx/nginx.conf test is successful
```

06 DNS Configuration — Namecheap

STEP 6.1 Point dineshtrade.online to EC2

1. Login to namecheap.com
2. Domain List → Manage → dineshtrade.online
3. Click Advanced DNS tab
4. Delete existing records (CNAME www and URL Redirect @)
5. Add two new A Records:

Type	Host	Value	TTL
A Record	@	3.111.255.172	Automatic
A Record	www	3.111.255.172	Automatic

DNS propagation takes 5-30 minutes. Test with: `ping dineshtrade.online` — should resolve to 3.111.255.172

07 SSL Certificate — Free HTTPS via Let's Encrypt

STEP 7.1 Install Certbot and Get SSL Certificate

```
# Install Certbot

sudo apt install -y certbot python3-certbot-nginx

# Get SSL certificate (DNS must be propagated first)

sudo certbot --nginx -d dineshtrade.online -d www.dineshtrade.online
```

```
# When prompted:

# Email: dinesh.k.wadhwani@gmail.com

# Agree to terms: A

# Share with EFF: Y or N (optional)


# Expected success message:

# Successfully received certificate.

# Congratulations! HTTPS enabled on https://dineshtrade.online
```

Certificate auto-renews every 90 days via systemd timer. Current certificate expires: 2026-08-16. No manual action needed for renewal.

STEP 8.1

First-Time Setup (Local Mac — VS Code Terminal)

```
# 1. Ensure .gitignore has these entries:

# .env.local

# node_modules/

# .next/

# *.pem


# 2. Initialise Git

cd ~/Downloads/dineshtrade

git init

git add .

git commit -m "Initial commit - DineshTrade v1.0"


# 3. Create private repo on github.com then:

git remote add origin https://github.com/YOUR_USERNAME/dineshtrade.git

git branch -M main

git push -u origin main
```

STEP 8.2

First-Time Setup (EC2 Server — SSH Terminal)

```
# Install Git

sudo apt install -y git


# Clone repository

cd ~

git clone https://github.com/YOUR_USERNAME/dineshtrade.git
```

```
cd dineshtrade

# Create .env.local (ONE TIME ONLY – never overwritten by deployments)

nano .env.local

# Paste all environment variables, save with Ctrl+X → Y → Enter

# Install dependencies and build

npm install

npm run build

# Start with PM2

pm2 start npm --name "dineshtrade" -- start

pm2 save

pm2 startup # follow the command it gives you

# Reboot server

sudo reboot

# After 30 seconds: open https://dineshtrade.online
```

STEP 8.3**Every Future Deployment (After Code Changes)****On your Mac (VS Code terminal):**

```
git add .

git commit -m "describe what you changed"

git push
```

Then SSH into server:

```
ssh -i ~/Downloads/dineshtrade-server.pem ubuntu@3.111.255.172

cd ~/dineshtrade
```

```
git pull
```

```
npm install
```

```
npm run build
```

```
pm2 restart dineshtrade
```

Total deployment time after initial setup: ~2-3 minutes. The .env.local file on the server is NEVER touched by deployments — your API keys are safe.

These go in .env.local on the SERVER (created once, never committed to Git):

```
# AI Provider (swappable – change 3 vars only)

AI_API_KEY=sk-ant-xxxxx # from console.anthropic.com

AI_API_URL=https://api.anthropic.com/v1/messages

AI_MODEL=claude-haiku-4-5-20251001


# Zerodha Kite Connect

ZERODHA_API_KEY=xxxxx # from developers.kite.trade

ZERODHA_API_SECRET=xxxxx # from developers.kite.trade

# NOTE: Access token entered daily via Settings page – NOT here


# Session Security

SESSION_SECRET=dineshtrade2026anyrandom32charstring


# App

NEXT_PUBLIC_APP_NAME=DineshTrade

NEXT_PUBLIC_APP_VERSION=1.0.0
```

SSH Access:

```
ssh -i ~/Downloads/dineshtrade-server.pem ubuntu@3.111.255.172
```

PM2 Commands:

```
pm2 status # check if app is running

pm2 logs dineshtrade # view live app logs

pm2 restart dineshtrade # restart after deployment
```

```
pm2 stop dineshtrade # stop the app

pm2 monit # live CPU/memory dashboard
```

Nginx Commands:

```
sudo nginx -t # test config syntax

sudo systemctl restart nginx # restart nginx

sudo systemctl status nginx # check nginx status

sudo tail -f /var/log/nginx/error.log # view error logs
```

SSL Certificate:

```
sudo certbot renew --dry-run # test auto-renewal

sudo certbot certificates # view cert expiry dates
```

Server Maintenance:

```
sudo reboot # reboot server

df -h # check disk space

free -h # check memory

htop # live process monitor (install: sudo apt install htop)
```

Setup Status Summary

Component	Status	Details
EC2 Instance	Running	t3.micro — Mumbai ap-south-1
Static IP	Assigned	3.111.255.172 (Elastic IP)
Domain	Configured	dineshtrade.online → 3.111.255.172
SSL	Active	Let's Encrypt — expires 2026-08-16
Node.js	Installed	v20.20.2
PM2	Installed	v7.0.1
Nginx	Running	v1.24.0
Zerodha IP	To Register	Register 3.111.255.172 at developers.kite.trade

App	Pending Deploy	Complete GitHub setup then deploy
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